



Property Field Mappings in CADTalk 3.0 for SYSPRO

From: SolutionsX, LLC.

This document contains information regarding property field mappings in CADTalk 3.0 for SYSPRO.



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Overview

At the heart of the CADTalk product are the property field mappings. To harness the inherent power, there is an expression builder. The expression builder is used to create expressions as simple as direct mappings (CAD Property Field to SYSPRO Field) or as complex as can be imagined with multiple IF/THEN statements or other operators.

To get the CAD Model's data into SYSPRO, these defined property field mappings are applied to the model when CADTalk performs its initial load of the model. This document defines the multiple objects, describes the order in which property field mappings are applied to the model, and gives multiple examples of common property field mappings.

Definitions

SYSPROStockItem – Represents a single row in the Stock Code Master table in SYSPRO and contains all of the Stock Code field data. i.e. Description, Product Class, Warehouse to Use.

***[CADPrefix]Item** – Represents data found in the CAD Model or Drawing. Typically, this would contain any data property that is independent of its occurrence in a particular assembly. i.e. Description, Mass Properties, Custom Properties.

SYSPROBOMStructure – Represents a single row in the SYSPRO BOM Structure table in SYSPRO.

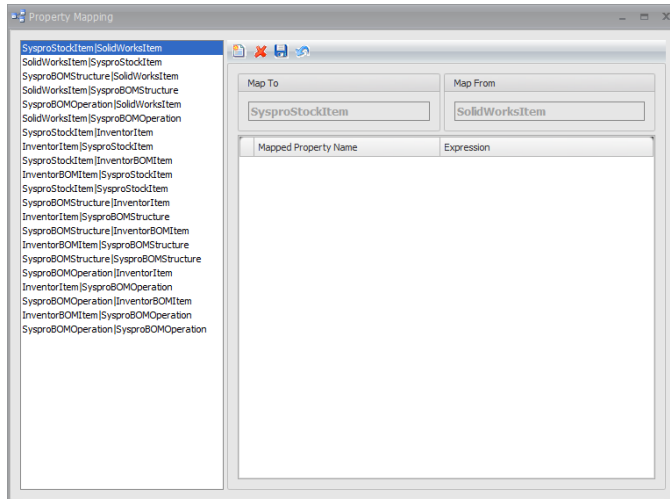
***[CADPrefix]BOMItem** – Represents an occurrence in the BOM of a CAD Model or drawing. This object typically has an associated CAD Item for reference but includes occurrence level data such as quantity and sequence numbering (balloon numbers).

SYSPROBOMOperation – Represents a single row in the SYSPRO BOM Operations table in Syspro.

*The CAD prefix would be the name of the particular CAD system. For example with Autodesk Inventor, this would be InventorItem, InventorBOMItem.

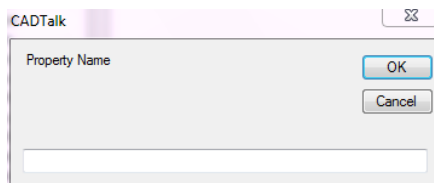
Data Field Mapping – How To

Open the Property Mapping Dialog box by clicking the Data Field Mappings icon in the Data group of the toolbar.

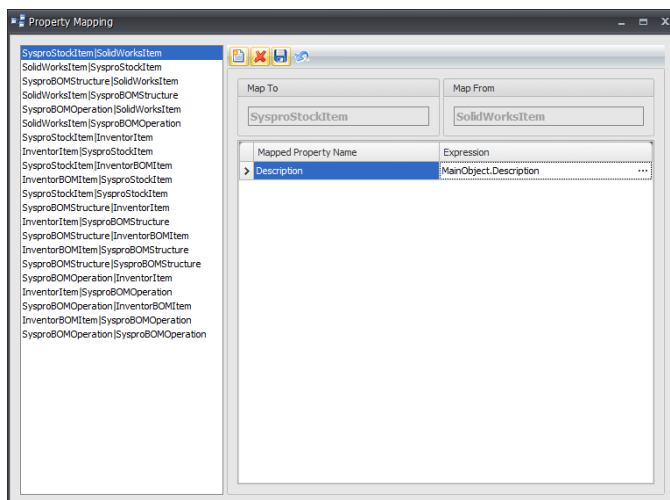


In the left pane, the available mapping end points are displayed. The endpoint listed first is the “Map To” and the endpoint listed second is the “Map From”. This is verified in the right pane.

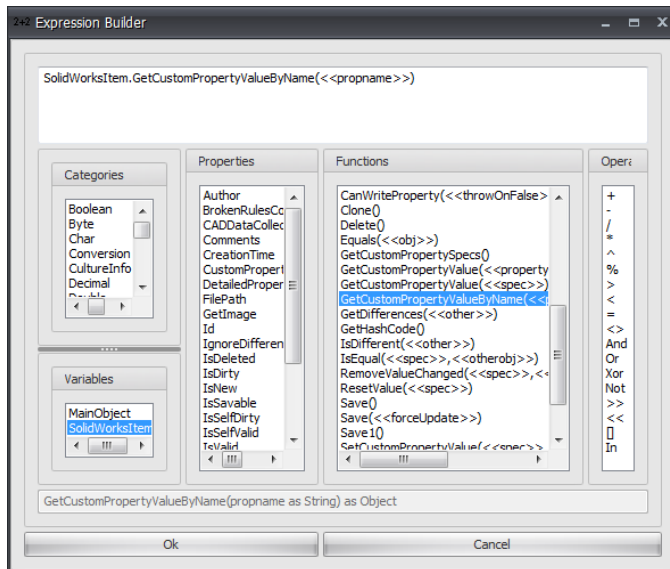
To create a mapping from a SolidworksItem to a SYSPRO Stock Item, click the New icon. 



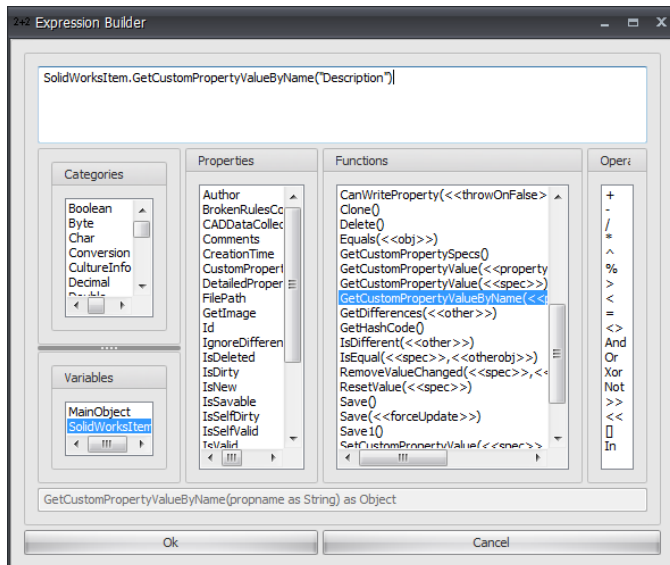
Type the name of the SYSPRO Field you want to populate with data from a CAD property. For this example, we’ll use Description. NOTE: On direct property mappings this Property Name MUST match the field name in the SYSPRO Stock Code Master Table. If the property name here cannot be found in the Stock Code Master Table then it is assumed that this is an action instead of a direct mapping.



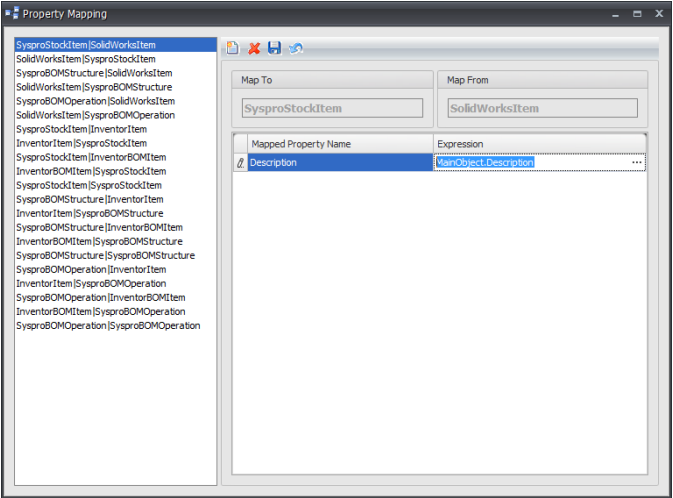
Click the three dots in the Expression field to open the expression builder. 



Update the <<proprname>> with the property name from the SolidWorks model; “Description” in this case. NOTE: The name of the SYSPRO field we are updating is Description so that is the Property Name specified earlier in this example AND the SolidWorks custom property happens to match although it certainly doesn’t have to.



Click Ok to close the Expression Builder.



Click the Save icon to save the mapping. 

Order of Processing

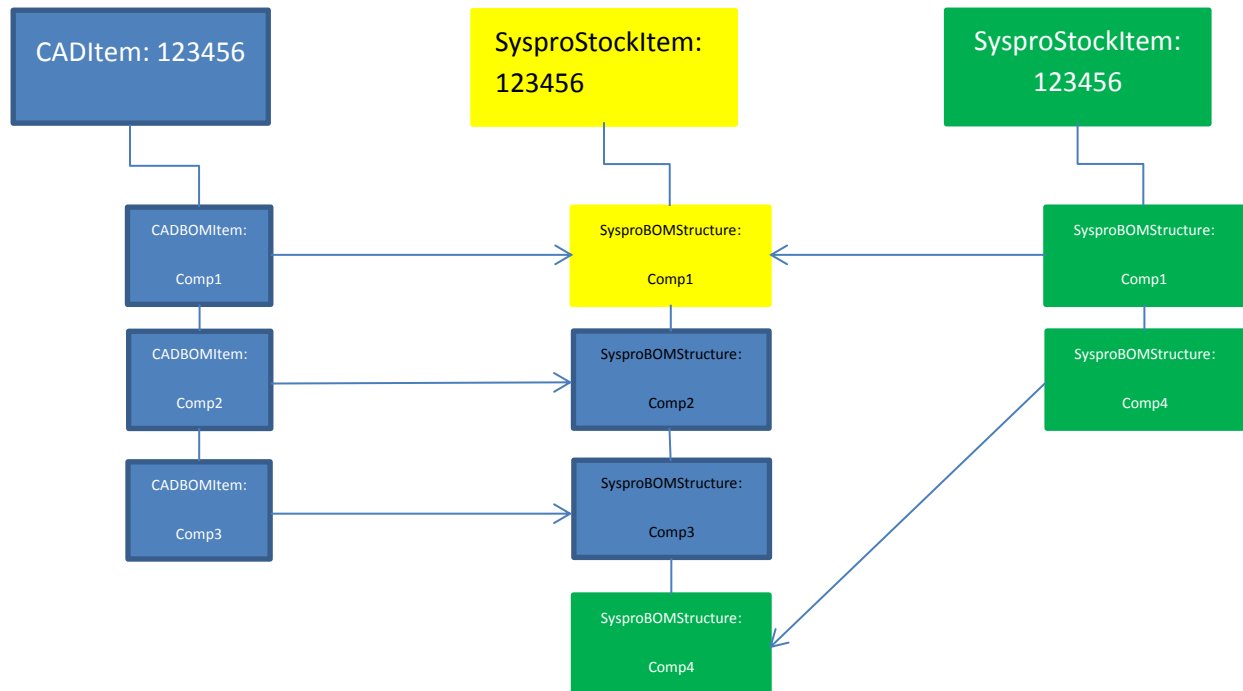
When a user opens a CAD file within CADTalk it will transform the CAD representation of the Stock Code/BOM into a proposed SYSPRO representation of the same Stock Code/BOM. This transformation is done in the following order. Note: mappings/actions within endpoints are executed in the same order that they are shown in the property mapping form.

1. Property Mappings from the CADItem to the SysproStockItem are executed against all Stock Codes (including the parent item) in the CAD Model.
 - a. The ID (Stock Code) from the CADItem is used to pull the existing SysproStockItem from SYSPRO. If the stock code doesn't exist, a SysproStockItem will be generated (within CADTalk) using the Stock Code defaults defined within SYSPRO.
 - i. Note: In order for CADTalk to pull existing stock codes from SYSPRO, the ID (stock code) represented by the incoming CADItem must be known prior to mapping. Therefore the Stock Code field cannot be mapped during property mapping. Settings related to the identification of the CADItem ID (Stock Code) are configured in the settings for each CAD system.
 - b. Once the SysproStockItem is retrieved from SYSPRO, each property mapping is executed using the CADItem and the SysproStockItem (MainObject in the mapping screen).
2. Property Mappings are executed from CADItem to SysproBOMOperation. CADTalk looks at each operation that was created in the first step (or already existing on the existing SysproStockItem) and applies any mappings to those operations. This is how users can use attributes from a CAD Item to update properties like Run Time on the operations. For example, the run time for a particular work center can be calculated based on the mass of the model or the length of a particular feature.
3. All components found in the CAD Model are conditionally merged with the existing BOM from SYSPRO if the item already exists. Please see BOM Merging for details.
4. Property Mappings from CADBOMItem are mapped to the SysproBOMStructure that was created in step 3. The CADBOMItem typically has properties that are specific to just that instance of the component in the assembly like Quantity and Item Number or Balloon Number.
5. Property Mappings from CADItem to SysproBOMStructure are executed for each component.

NOTE: There are predefined mappings such as SYSPROStockItem to SYSPROStockItem that are used to control which properties are copied when copying from an existing template item already in SYSPRO using the TemplateStockCode property in CADTalk. These mappings are invoked when TemplateStockCode field on a SysproStockItem is changed within CADTalk after having provided CADTalk with a Stock Code value in that field. Typically, the TemplateStockCode field would be populated during property mapping in order to copy default field values from the existing template Stock Code in SYSPRO as well as operations. It is advisable to make the TemplateStockCode mapping the first mapping to allow overriding of specific properties in subsequent mappings.

BOM Merging

CADTalk looks for existing SysproBOMStructure with the same stock code as the CADBOMItem in the existing SysproBOMStructure. If a structure doesn't exist, a blank new structure is created for mapping. CADTalk then executes the mappings for CADBOMItem to SysproBOMStructure. Note that existing structures in the existing SysproStockItem are retained during the merge. After property mapping is executed, the IncludeInBOM property on the SysproBOMStructure is checked prior to adding the structure to the new merged BOM.



Example Data Field Mapping Expressions

For these examples, “CADItem” is utilized for norming. The actual titles will depend on the CAD system utilized in the user’s environment.

Straight Mappings from Custom CAD Properties

Map from: *CADItem*

Map To: *SysproStockItem*

Mapped Property Name/Action Description: *Description*

Expression:

CADItem.GetCustomPropertyValueByName("Description") where “Description” is the label of the custom property on the CADItem.

Trim Data to a Specific Length

Map from: *CADItem*

Map To: *SysproStockItem*

Mapped Property Name/Action Description: *Description*

Expression:

Left(CADItem.GetCustomPropertyValueByName("Description"),30)

Compare Data with Existing SYSPRO Data

Map from: *CADItem*

Map To: *SysproStockItem*

Mapped Property Name/Action Description: *ProductClass*

Expression:

If(MainObject.IsNew OR (String.Compare(MainObject.Release,CADItem.RevisionNumber,True) < 0),CADItem.CostCenter,MainObject.ProductClass)

NOTE: This expression evaluates two different situations: 1.) Does the item exist in SYSPRO? 2.) If it is an existing SYSPRO item, is the revision number in the CAD system greater than the release field in Syspro ? If the answer is yes to either of these questions, then populate the Product Class with data in the CAD Item’s property labeled “CostCenter”; if not, then use the Product Class of the existing SYSPRO item.

Create Component Material Based on CADItem Properties and Specify Quantity Per

Map from: *CADItem*

Map To: *SysproStockItem*

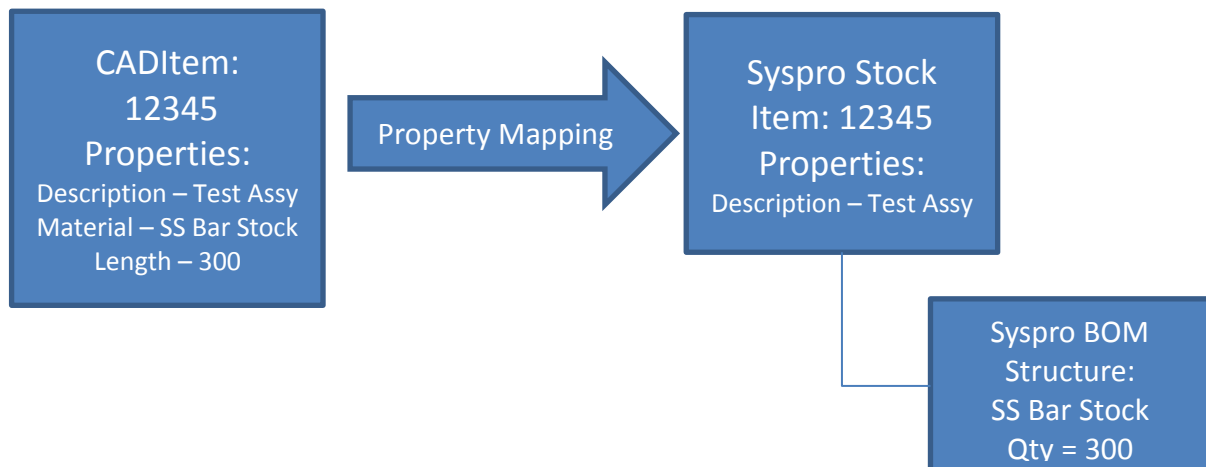
Mapped Property Name/Action Description: *CreateMaterial*

Note: In this case the Mapped Property Name would not match an actual SYSPRO field name as we are performing an action of adding structure to the SysproStockItem instead of mapping to a particular field.

Expression:

MainObject.AddMaterialToFirstOperation(CADItem.StockNumber,Decimal.Parse(InventorItem.GetCustomPropertyValueByName("Length")))

This example shows how properties within the CAD model/drawing can be used to create components in Syspro. Typically, raw material for a particular item is represented in the CAD system as a property on the model. However, in Syspro this needs to be represented as a BOM Structure.



Evaluate BOM Item for Inclusion as a Component

Map from: *CADBOMItem*

Map To: *SysproBOMStructure*

Mapped Property Name/Action Description: *IncludeInBom*

If((CADItem.DefaultBOMStructure <> "Normal" AND CADItem.DefaultBOMStructure <> "Phantom" AND CADItem.DefaultBOMStructure <> "Purchased") or (CADItem.PartNumber = "" and CADItem.StockNumber = ""), False, True)